Simulation of the Noisy Harmonic Oscillator: Gaussian and Lévy White Noise Forcing

Tristan Mihocko

May 2026

Abstract

The harmonic oscillator is one of the most fundamental models in applied mathematics, describing physical systems from mechanical springs and electrical circuits to molecular vibrations. In idealized settings the system evolves deterministically, but in practice external forces are often random: thermal fluctuations, turbulent flow, or unpredictable impulses perturb the trajectory continuously. Replacing the deterministic forcing with white noise converts the governing ordinary differential equation into a stochastic differential equation (SDE), whose solutions are random processes rather than fixed curves.

This project presents an interactive simulation of the noisy harmonic oscillator driven by either Gaussian white noise or Lévy stable white noise. The simulation numerically integrates the SDE using the Euler–Maruyama method and produces two plots: the displacement time series $x(t)$ and the phase portrait in the (x, v) plane. Both plots update in real time through interactive `matplotlib` sliders, allowing the user to explore how the physical parameters and the character of the noise shape the system's behavior. The tool also compares the qualitatively different trajectories produced by Gaussian and Lévy noise, illustrating the role of the stability index α in generating heavy-tailed, jump-dominated dynamics.

1 Mathematical Background

This section develops the mathematical framework underlying the simulation, beginning with the classical oscillator and building up to the stochastic setting and the two noise models.

1.1 The Classical Damped Harmonic Oscillator

The damped harmonic oscillator is governed by the second-order linear ODE:

$$m\,x''(t) + c\,x'(t) + k\,x(t) = 0 \tag{1}$$

where $x(t)$ is the displacement from equilibrium, $m > 0$ is the mass, $c \geq 0$ is the damping coefficient, and $k > 0$ is the spring constant. Two derived values characterize the system's behavior:

the natural frequency at which the undamped system oscillates (2), and the critical damping value (3), which separates oscillatory from non-oscillatory behavior.

$$\omega_0 = \sqrt{k/m} \tag{2}$$

$$c_{\text{crit}} = 2\sqrt{km} \tag{3}$$

The qualitative character of the solution depends on the ratio $\zeta = c/c_{\text{crit}}$:

- **Underdamped** ($c < c_{\text{crit}}$): the system oscillates at frequency $\omega_d = \omega_0\sqrt{1-\zeta^2}$ with amplitude decaying as $e^{-\zeta\omega_0 t}$. In the phase portrait this appears as an inward spiral converging to the origin.

- **Critically damped** ($c = c_{\text{crit}}$): the system returns to equilibrium as quickly as possible without oscillating. The phase portrait shows a trajectory curving directly into the origin.

- **Overdamped** ($c > c_{\text{crit}}$): the system returns to equilibrium slowly without oscillating, as a sum of two decaying exponentials. The phase portrait shows a trajectory that approaches the origin along a nearly straight path.

1.2 The Phase Portrait

The phase portrait is a plot of the velocity $v(t) = x'(t)$ and the displacement $x(t)$, tracing the trajectory of the system in the two-dimensional space (x, v). Rather than showing how x or v evolve over time individually, the phase portrait shows the relationship between position and velocity across the entire trajectory, making it easier for the human eye to notice qualitative features.

For the unforced damped harmonic oscillator, the phase portrait is a smooth spiral (underdamped) or a curve approaching the origin (overdamped or critically damped). Adding noise disrupts this smooth structure: the trajectory wanders randomly through the (x, v) plane, tracing an irregular, space-filling curve rather than a clean spiral. Under Lévy noise, sudden large increments in v produce sharp, nearly vertical, jumps in the phase portrait, which are very distinct from the gradual drift seen under Gaussian noise.

1.3 Stochastic Processes

A stochastic process is a collection of random variables $\{X_t\}_{t\geq 0}$ indexed by time, representing a probability distribution over infinitely many possible trajectories called *sample paths*. A single run of the simulation produces one such path. Re-running the simulation with the same parameters produces a different path with the same statistical properties. The randomness is not a numerical error; it is a core feature of the model.

1.4 Stochastic Differential Equations

While an ODE describes a system whose future state is completely determined by its current state via deterministic rules, an SDE introduces randomness directly into those rules by adding a noise term which is a stochastic process. Consequently, the solution of an SDE is not a single curve but a stochastic process.

To add stochastic forcing to the classic damped harmonic oscillator (1), we replace the right-hand side with a white noise term $F(t)$:

$$m\,x''(t) + c\,x'(t) + k\,x(t) = F(t) \tag{4}$$

Introducing the velocity $v(t) = x'(t)$ reduces this to a first-order system, which is the standard form for numerical integration:

$$dx = v\,dt \tag{5}$$

$$dv = \left(-\frac{c}{m}\,v - \frac{k}{m}\,x\right)dt + \frac{1}{m}\,dW_t \tag{6}$$

where dW_t is the differential noise increment over the time step dt. Its distribution depends on the chosen noise model.

1.5 Gaussian White Noise

The standard model of continuous random forcing is the Wiener process (Brownian motion) W_t. Its increments $dW_t = W_{t+dt} - W_t$ are:

- independent for non-overlapping time intervals,
- normally distributed: $dW_t \sim \mathcal{N}(0,\,\sigma^2 dt)$,
- continuous almost surely.

The parameter $\sigma > 0$ is the noise scale. Because the variance of each increment is proportional to dt, the typical magnitude of a fluctuation over a small step scales like $\sigma\sqrt{dt}$, which is much larger than the drift contribution, which scales linearly with dt, for small Δt. This is what causes the famous roughness of Brownian paths. The resulting SDE is treated in the framework of Itô calculus, with Itô's lemma providing the fundamental rules for differentiation and analysis of stochastic processes.

1.6 Lévy Stable White Noise

A more general family of noise processes is built from the Lévy stable distributions. A random variable X follows the stable distribution $S(\alpha, \beta, \sigma, \mu)$ characterized by four parameters:

- $\alpha \in (0, 2]$: the **stability index**. Controls how heavy the tails of the distribution are. When $\alpha = 2$ the distribution is exactly Gaussian. When $\alpha < 2$ the tails of the distribution satisfy $f(x) \sim |x|^{-(\alpha+1)}$, meaning the distribution has infinite variance and the noise can produce arbitrarily large increments with a non-negligible probability.
- $\beta \in [-1, 1]$: the **skewness parameter**. When $\beta = 0$ the distribution is symmetric. When $\beta = 1$ it is completely right-skewed and when $\beta = -1$ completely left-skewed.
- $\sigma > 0$: the **scale parameter**, controlling the spread of the distribution.
- $\mu \in \mathbb{R}$: the **location parameter** shifts the distribution along the real line. It is standard practice to set $\mu = 0$, centering noise about the origin.

Lévy stable distributions are defined through their characteristic function rather than a closed-form density.

$$\log \mathbb{E}\left[e^{itX}\right] = -\sigma^{\alpha}|t|^{\alpha}(1 - i\beta \operatorname{sgn}(t)\Phi(\alpha, t)) + i\mu t \tag{7}$$

$$\Phi(\alpha, t) = \begin{cases} \tan\left(\frac{\pi\alpha}{2}\right) & \text{if } \alpha \neq 1 \\ -\frac{2}{\pi}\ln|t| & \text{if } \alpha = 1 \end{cases} \tag{8}$$

The Lévy stable noise increment over a step Δt is drawn from:

$$\Delta W \sim S\big(\alpha,\, \beta,\, \sigma\, \Delta t^{1/\alpha},\, 0\big) \tag{9}$$

The $\Delta t^{1/\alpha}$ scaling is the natural generalization of the $\sqrt{\Delta t} = \Delta t^{1/2}$ scaling of Gaussian increments, recovering the Gaussian case when $\alpha = 2$. For $\alpha < 2$ the distribution has heavier tails than the Gaussian, producing occasional large jumps in $x(t)$ and sharp excursions in the phase portrait. This makes Lévy noise a natural model for systems subject to rare but extreme shocks.

2 Numerical Method

2.1 The Euler–Maruyama Scheme

The Euler–Maruyama method is the SDE analog of the famous Euler method for ODEs. Applied to the system (5)–(6) with time step Δt, it gives the update:

$$x_{i+1} = x_i + v_i\,\Delta t \tag{10}$$

$$v_{i+1} = v_i + \left(-\frac{c}{m}\,v_i - \frac{k}{m}\,x_i\right)\Delta t + \frac{\Delta W_i}{m} \tag{11}$$

where ΔW_i is a single random draw from the noise distribution. For Gaussian noise, $\Delta W_i \sim \mathcal{N}(0, \sigma^2 \Delta t)$. For Lévy noise, $\Delta W_i \sim S(\alpha, \beta, \sigma\,\Delta t^{1/\alpha}, 0)$.

Equation (10) advances the position using the current velocity, and equation (11) advances the velocity using the current deterministic drift plus a random noise kick. The key difference from the ODE Euler method is the ΔW_i term, which injects randomness at every step.

2.2 Convergence

For SDEs driven by Wiener processes, the Euler–Maruyama method has strong convergence of order 1/2: the expected absolute error $\mathbb{E}[|x(T) - x_N|]$ scales as $O(\sqrt{\Delta t})$. This is slower than the $O(\Delta t)$ convergence of forward Euler for ODEs, a consequence of the nowhere-differentiability of Brownian paths. For Lévy-driven SDEs with $\alpha < 2$ the convergence theory is more involved due to the infinite variance of the increments; in practice the step size $\Delta t = 0.01$ produces visually stable trajectories for the parameter ranges used here.

2.3 Numerical Parameters

Parameter	**Symbol**	**Value**
Mass	m	1.0 kg
Time step	Δt	0.01 s
Total time	T	40.0 s
Number of steps	N	4000
Initial displacement	$x(0)$	0.5 m
Initial velocity	$v(0)$	0.0 m/s

Table 1: Fixed numerical parameters used in all simulations.

3 Implementation

The simulation is implemented in Python using `numpy`, `scipy`, and `matplotlib`.

3.1 Noise Sampling

At each time step a noise increment is drawn from the selected distribution. For Gaussian noise, `numpy.random.normal(0, sigma * sqrt(dt))` is used. For Lévy stable noise, `scipy.stats.levy_stable.rv` is called with the user-specified α and β and scale $\sigma\,\Delta t^{1/\alpha}$.

3.2 Visualization

The figure contains two panels that update together whenever any parameter is changed.

Left panel — displacement $x(t)$**:** plots the displacement against time over the full 40 s simulation window. This shows the oscillatory structure of the trajectory, the effect of damping on amplitude, and the character of the noise (smooth fluctuations for Gaussian, intermittent jumps for Lévy).

Right panel — phase portrait: plots the velocity $v(t)$ against the displacement $x(t)$ as a parametric curve, tracing the system's trajectory. For the unforced underdamped oscillator this is a smooth inward spiral. Adding noise disrupts this into an irregular, wandering curve. The white marker indicates the starting point $(x(0), v(0)) = (0.5, 0)$.

3.3 Interactive Controls

Five `Slider` widgets allow real-time adjustment of the simulation parameters:

Slider	**Parameter**	**Range**	**Default**
Damping c	Damping coefficient	$[0.01,\ 3.0]$	0.20
Spring k	Spring constant	$[0.50,\ 12.0]$	4.00
Noise scale σ	Noise intensity	$[0.01,\ 2.0]$	0.30
Lévy α	Stability index	$[0.50,\ 2.0]$	1.50
Lévy β	Skewness	$[-1.0,\ 1.0]$	0.00

Table 2: Interactive sliders, their parameters, ranges, and default values.

A `RadioButtons` widget switches between Lévy stable and Gaussian noise. A **Re-run** button generates a new noise path at the current parameter values. A text panel below the controls displays the natural frequency f_0, critical damping c_{crit}, and current damping regime, updating automatically as the sliders are moved.

4 Results

4.1 Unforced System ($\sigma \approx 0$)

Setting the noise scale near zero yields approximately deterministic behavior. With $c = 0.2$ and $k = 4.0$ (underdamped, $c_{\text{crit}} = 4.0$), the $x(t)$ plot shows an oscillation at $f_0 \approx 0.318$ Hz whose amplitude decays exponentially. The phase portrait traces a clean inward spiral converging to the origin. Increasing c past c_{crit} eliminates the oscillation: $x(t)$ decreases monotonically and the phase portrait shows a curve approaching the origin along an almost straight path.

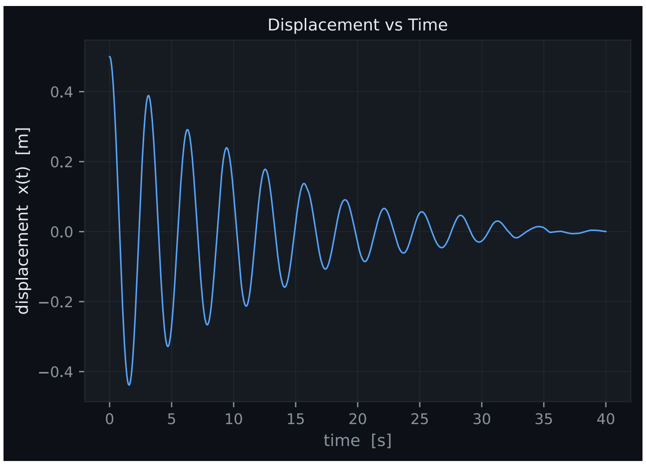

Figure 1: Displacement of oscillator under Lévy noise with noise scale $\sigma \approx 0$

Figure 2: Phase portrait of oscillator under Lévy noise with noise scale $\sigma \approx 0$

4.2 Gaussian Noise

Under Gaussian forcing with $\sigma = 0.3$, the $x(t)$ plot shows sustained oscillations near ω_0 coupled with small random perturbations. The amplitude no longer decays to zero. Instead, the noise continuously re-excites the oscillator. In the phase portrait the clean spiral is replaced by a fuzzy, irregular band tracing approximately the same elliptical region, centered on the origin. Increasing σ widens this band; increasing c shrinks it by suppressing the oscillations. Different noise realizations at the same parameter values look broadly similar, reflecting the finite variance of the Gaussian distribution.

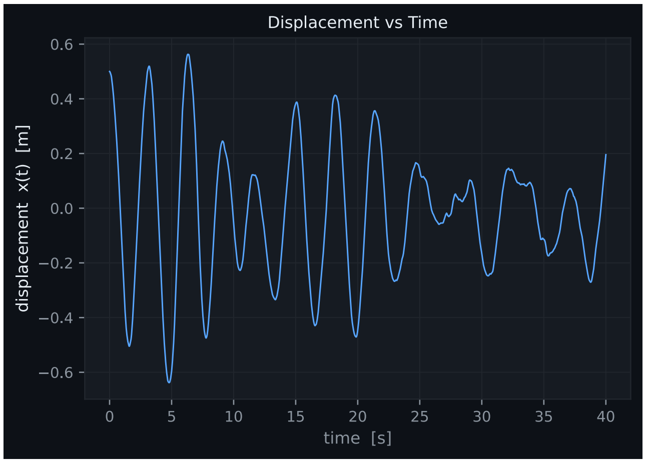

Figure 3: Displacement of oscillator under Gaussian noise

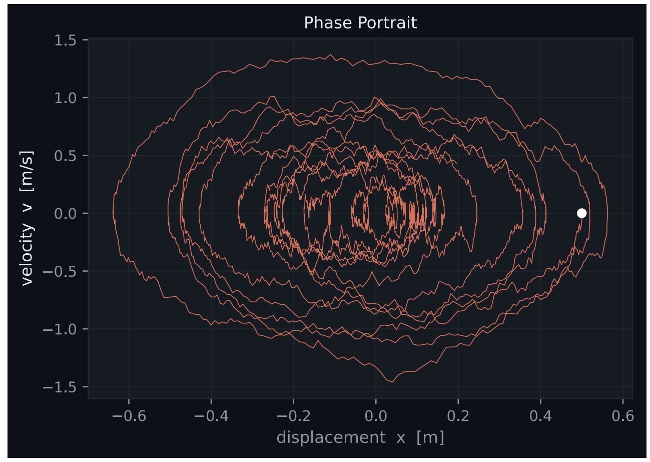

Figure 4: Phase portrait of oscillator under Gaussian noise

4.3 Lévy Noise

Switching to Lévy stable noise with $\alpha = 1.5$, $\beta = 0$, and $\sigma = 0.3$ produces qualitatively different behavior in both plots. The $x(t)$ time series is punctuated by intermittent large-amplitude jumps entirely absent under Gaussian noise, where the displacement suddenly shifts to a value far from its current level before the oscillator gradually rings back. In the phase portrait, these events appear as sudden near-vertical excursions in the v direction, representing instantaneous large velocity kicks, followed by curved segments as the system oscillates back.

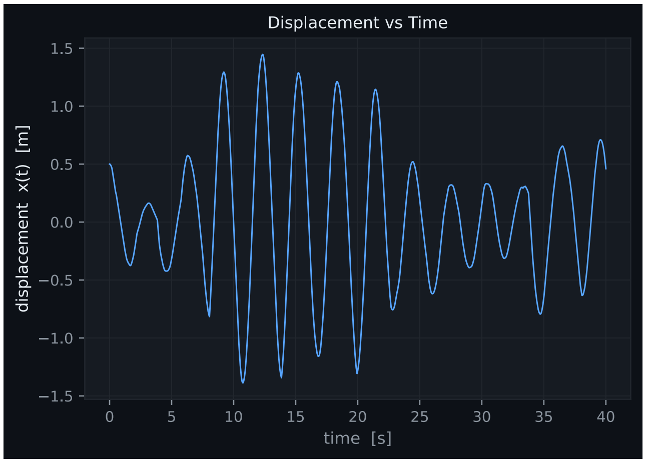

Figure 5: Displacement of oscillator under Lévy noise

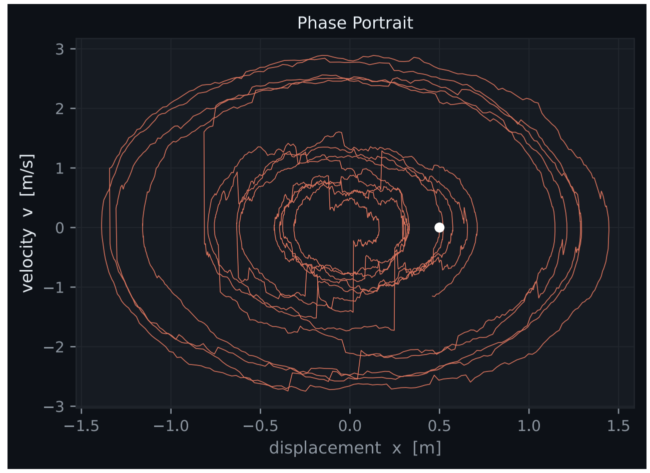

Figure 6: Phase portrait of oscillator under Lévy noise

Reducing α toward 0.5 increases both the frequency and magnitude of jumps. At $\alpha = 1.0$ the trajectory is dominated by jump events and the underlying oscillatory structure is barely visible between them. The skewness β controls the direction of the jumps: at $\beta = 1$ the jumps are predominantly in the positive x direction, which is visible in the phase portrait as asymmetric excursions above and to the right of the origin.

4.4 Comparison of Noise Models

The most direct way to compare the models is to switch between them using the radio buttons at identical parameter values. The key qualitative differences are summarized in the table below.

Feature	**Gaussian** ($\alpha = 2$)	**Lévy** ($\alpha < 2$)
Increment distribution	Finite variance	Infinite variance
Trajectory character	Smooth, small fluctuations	Intermittent large jumps
Phase portrait	Fuzzy band / irregular spiral	Spiral with sharp excursions
Path variability	Low (paths look similar)	High (jumps vary between runs)

Table 3: Qualitative comparison of Gaussian and Lévy noise models.

In both cases the oscillator continues to respond predominantly near the natural frequency $\omega_0 = \sqrt{k/m}$ between noise events, demonstrating that the characteristic response of the system is ignorant of the tail behavior of the noise.

5 Discussion

5.1 Strengths of the Tool

The simulation effectively illustrates the mathematical distinction between Gaussian and Lévy white noise. A user can slide α continuously from 2.0 downward and watch the trajectory transform from smooth random oscillation to jump-dominated behavior, making the role of the stability index immediately apparent. The two-panel layout is particularly useful because the $x(t)$ plot and the phase portrait highlight complementary aspects of the same trajectory: the time series shows when jumps occur, while the phase portrait shows the direction and magnitude of the resulting excursions. Together they give a more complete picture of the system's behavior than either plot alone.

The Re-run button adds interactivity that a static figure cannot provide. Because the solution is stochastic, pressing Re-run at fixed parameters shows a different realization each time, allowing the user to build intuition about variability: under Gaussian noise successive runs look nearly identical, while under Lévy noise the number, timing, and size of jumps can differ substantially from run to run.

5.2 Limitations

Numerical stability at small α. The Euler–Maruyama method can produce unreliable results when individual noise increments are very large relative to the current state, which occurs frequently at small α and large σ. In these regimes the velocity update adds an increment that can dwarf the drift term, occasionally causing the trajectory to diverge in a single step. A smaller time step or an implicit stochastic integrator would improve robustness in these extreme parameter regimes.

Single realization. The tool only displays one sample path at a time. Some useful statistics such as the mean displacement $\mathbb{E}[x(t)]$, the variance $\mathrm{Var}(x(t))$, or the marginal distribution of x require averaging over many realizations, which a single path cannot show.

Scope. The simulation is limited to a linear oscillator with scalar additive noise. Extensions would include nonlinear restoring forces (Van der Pol or Duffing oscillator), multiplicative noise where the noise amplitude depends on the state, or the display of additional diagnostics such as the power spectral density, which would reveal the peak at the natural frequency f_0 directly.